

Lecture 4

CMOS Inverter & Layout Design

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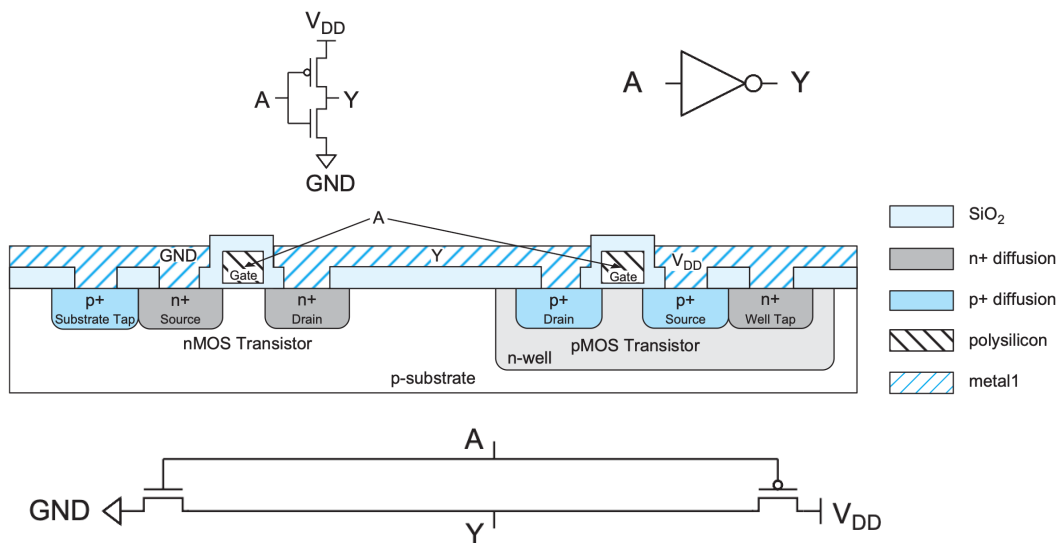


URL: <https://elec70142-digital-vlsi-design.github.io/course-website/>
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In this lecture, we focus on how an inverter works in some details. Further you will also learn:

- How to create the mask layer by designing the layout of an inverter.
- What are design rules and why they are required.
- How to design a good layout for a given transistor level circuit.
- What is the intended learning outcome from Lab 2.

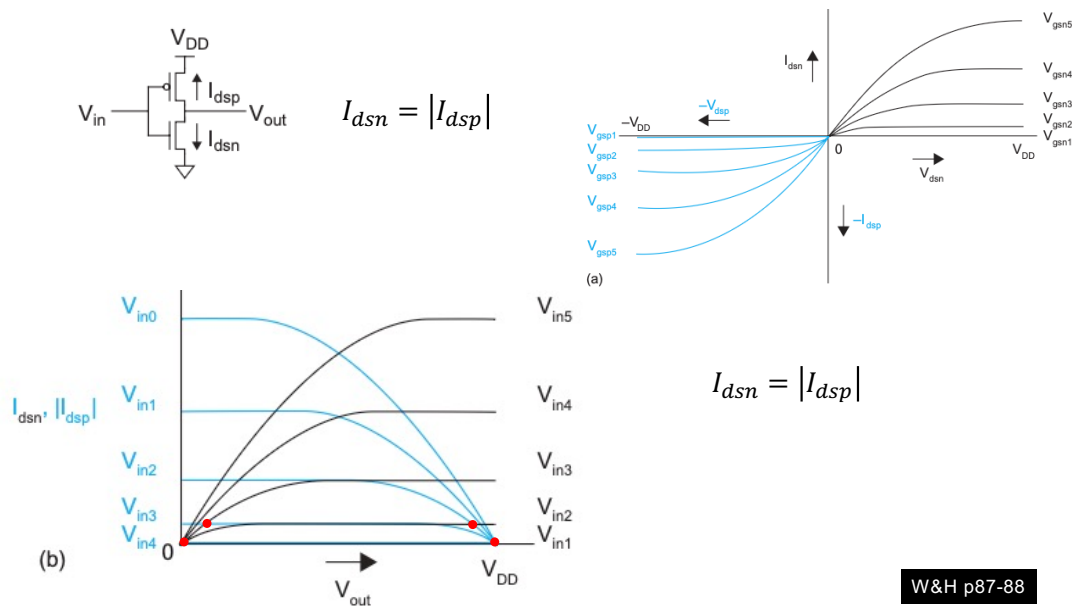
Cross section of an Inverter



All CMOS logic gates, no matter how complex, resemble an inverter. They have a pull-up path to VDD through pMOS transistors, and a pull-down path to VSS (GND) through nMOS transistors.

Here is the cross-section of an inverter and its corresponding transistor schematic.

Inverter DC characteristics



W&H p87-88

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EE4 – Digital VLSI Design

Lecture 4 Slide 3

Consider a simple inverter

When $V_{in} = V_{SS}$

$V_{out} = V_{DD}$

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$V_{out} = V_{SS}$

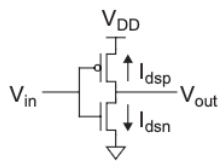
In between, V_{out} depends on current through transistors as determined by transistor width and length.

By KCL, steady state condition is:

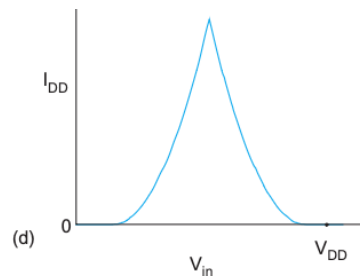
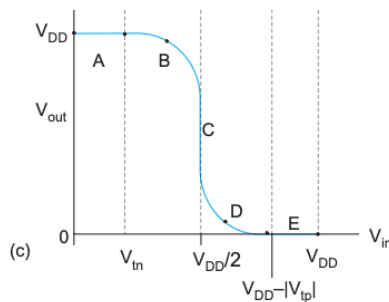
$$I_{dsn} = |I_{dsp}|$$

One can derive transfer function by solving equations. Here we use graphical method to gain some insights into its working.

Inverter switching threshold



Region	Condition	p-device	n-device	Output
A	$0 \leq V_{in} < V_{tn}$	linear	cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \leq V_{in} < V_{DD}/2$	linear	saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	saturated	saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{tp} $	saturated	linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - V_{tp} $	cutoff	linear	$V_{out} = 0$



W&H p88-89

The operation of the CMOS inverter can be divided into five regions from A to E as shown.

Region A: nMOS is OFF, pMOS pulls output to VDD.

Region B: the nMOS starts to turn ON, pulling the output down.

Region C: both transistors are in saturation. With ideal transistors, this happens when $V_{in} = V_{DD}/2$ and the transition happens immediately. However, due to finite output resistance of transistors, region C is broader.

Region D: pMOS is partially ON, but nMOS is in linear region.

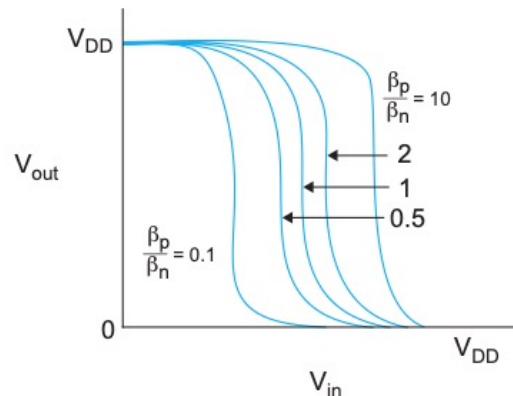
Region E: pMOS is completely OFF, nMOS pull the output down to VSS.

Also notice that the inverter's current consumption is ideally zero, neglecting leakage, when the input is within a threshold voltage of the VDD or GND rails. This feature is important for low-power operation.

In other words, the inverter only draws current through the supply rails DURING SWITCHING when both pMOS and nMOS transistors are conducting.

Beta ratio vs switching threshold

- ◆ Extract switching point depends on β_p / β_n
- ◆ If $\beta_p / \beta_n = 1$, switching occurs at around $V_{DD}/2$



W&H p90

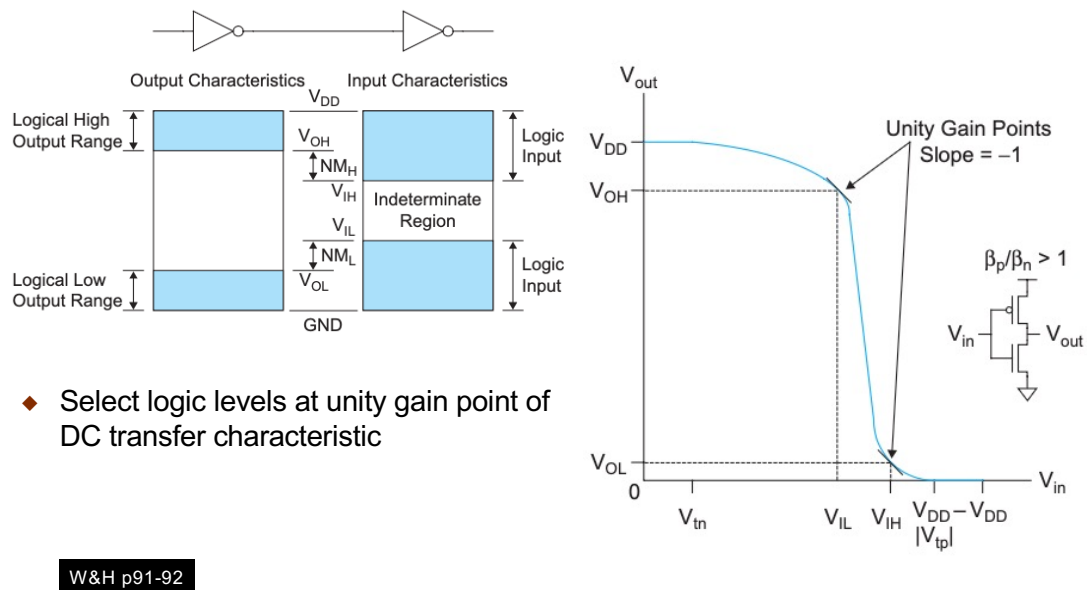
In general, one would design a gate such that the gain of the pull-up pMOS β_p and the pull-down nMOS β_n are the same, i.e. the **gain ratio** $r = \beta_p / \beta_n = 1$. As a result, the inverter switch threshold voltage will be $V_{DD}/2$, which is optimal from the noise margin point of view (discuss later). Equations 2.54 to 2.57 on W&H p90 shows that the switching threshold voltage for an inverter of gain ratio r is:

$$V_{inv} = \frac{V_{DD} + V_{tp} + V_{tn} \frac{1}{r}}{1 + \frac{1}{r}}$$

Since it $V_{tp} = -V_{tn}$, the switch threshold for $r=1$ is $V_{DD}/2$.

However, it is not always possible to achieve this ratio. The plot above show how the gain ratio affects the switching threshold. The dependency is not very strong.

Noise Margin in digital circuits



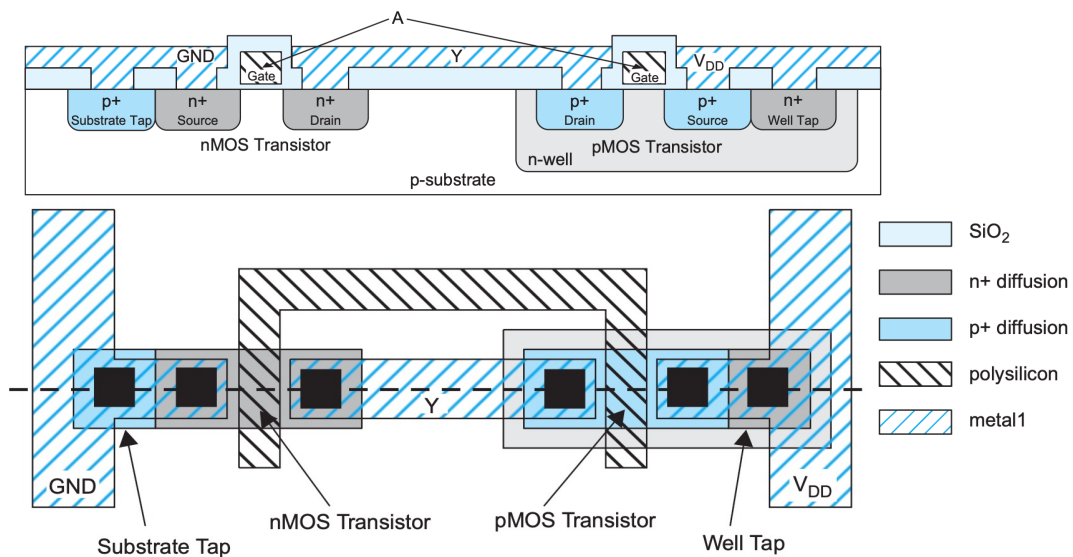
- ◆ Select logic levels at unity gain point of DC transfer characteristic

Noise margin is the allowable noise voltage on the input of a gate so that the output will not be corrupted. In the diagram above, the noise margin is the gaps between the two blue shade voltage level between output and input. The key voltages are:

- V_{IH} = minimum HIGH input voltage
- V_{IL} = maximum LOW input voltage
- V_{OH} = minimum HIGH output voltage
- V_{OL} = maximum LOW output voltage

These voltages are defined by the DC characteristic where the gradient is -1. When the gain ratio $r = 1$, the noise margin voltages are largest.

Cross-section & Layout of Inverter



W&H p21

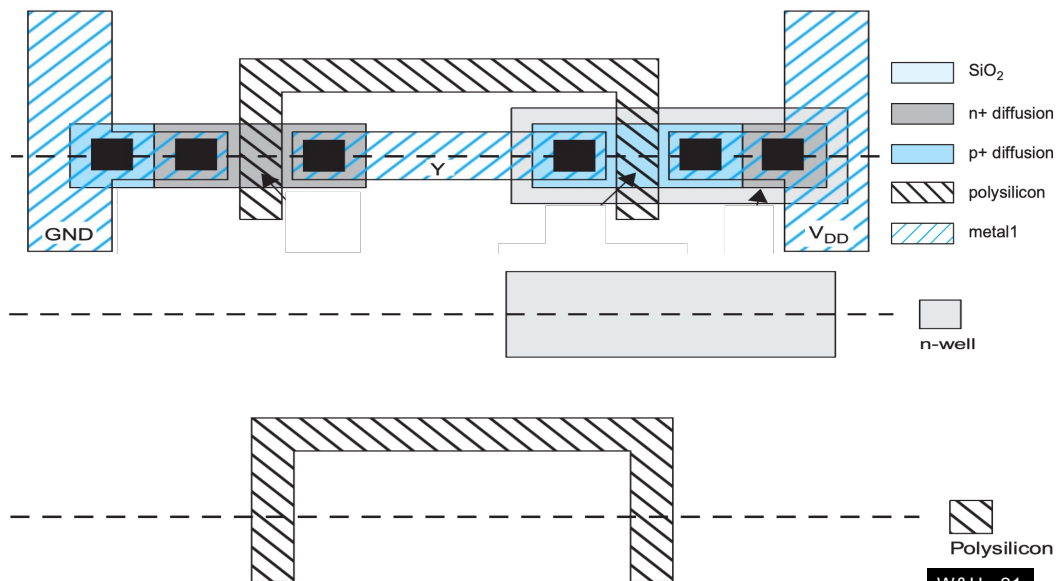
Fabrication uses a set of layers that are usually not exactly the same as the layout layers that the designer use because:

- It is easy to combine some layers together - for example, all “active” layers are diffusion layers (n+ and p+ diffusion) merged together.
- Other layers used for fabrication can be deduced from other layers. For example, the n-well region can be deduced from the p-type diffusion.
- Therefore, most layout CAD tools use mask layers that are more intuitive to the layout designer, and map to the real mask later.

CAD tools are used to automatically generate the real mask layers for fabrication

- Generating the manufacturing mask data is called ‘**tapeout**’
- Some geometrical layout rules are created to make sure that this tapeout process is possible.

Mask layout design of an Inverter (1)



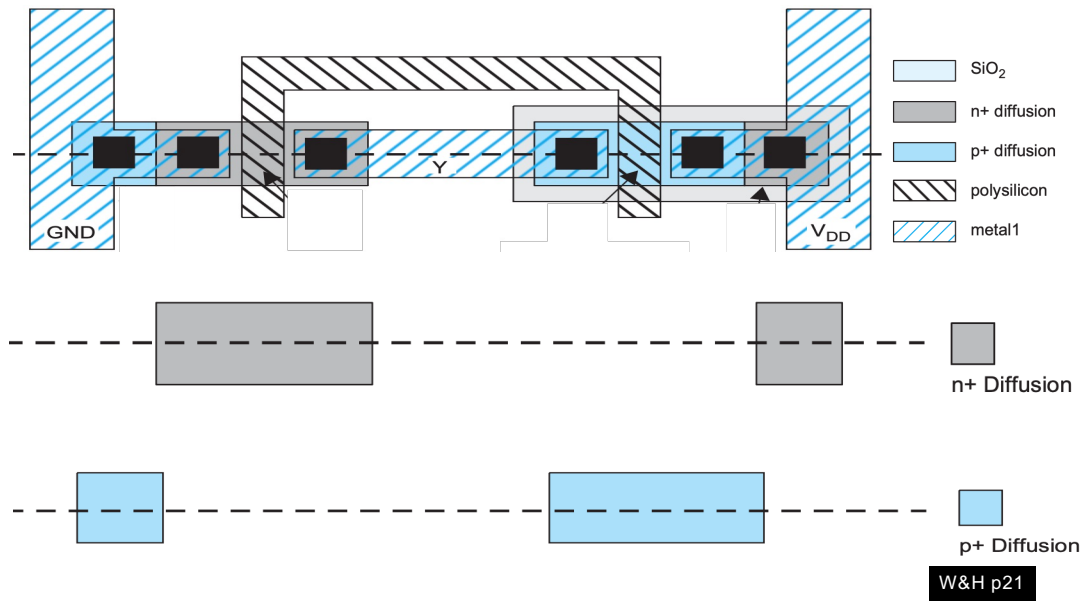
The next few slides show the different layers that a layout design would draw on a layout editor to specify an inverter.

Firstly, the n-well region is defined. Note that during tapeout, n-well regions may be automatically merged.

Not shown here is the active mask, which defines both the p and n diffusion regions that forms the source and drain of transistors. Active mask is often automatically generated so that the layout designer does not need to explicitly specify it.

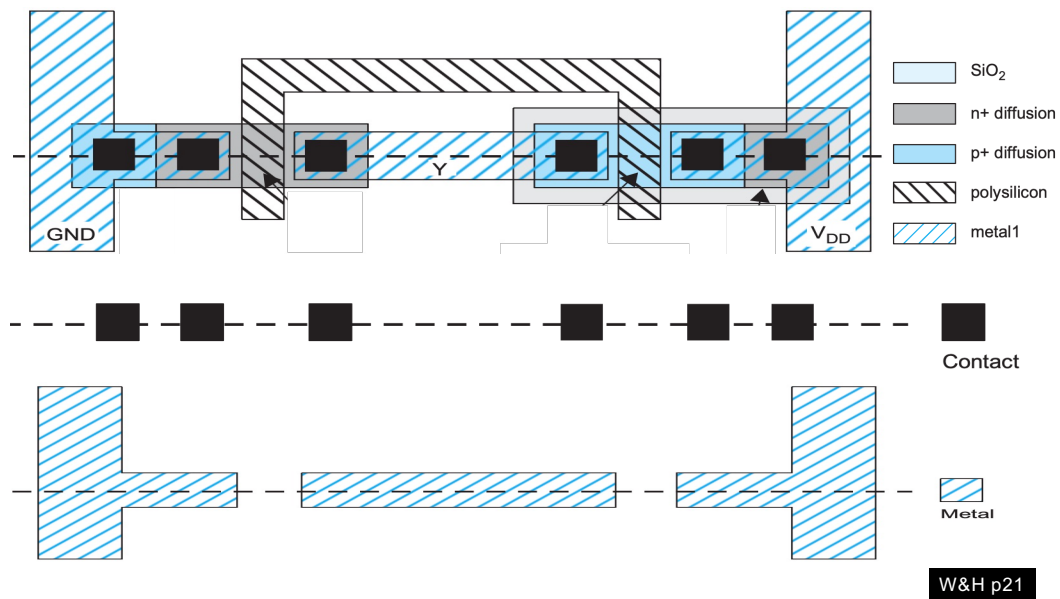
Next comes the polysilicon gate which will be used to fabricate the transistor such that the source and drain are self-aligned with the gate. The polysilicon gate layer will form the mask for the diffusion regions.

Mask layout design of an Inverter (2)



The next are the n+ and p+ diffusion masks. The “+” symbol here indicates that these regions are heavily doped with n or p type dopants. (n+ dopants have excess electrons and p+ dopants have excess holes, or missing electrons).

Mask layout design of an Inverter (3)

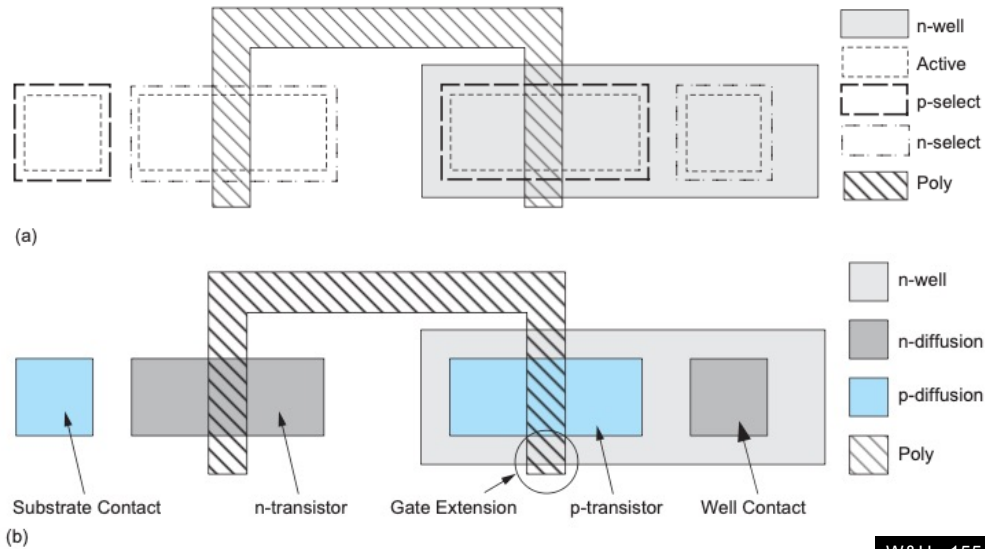


Next comes the contact holes. These are locations where vias are formed. Vias are feedthroughs between two layers of materials.

A via is made up of the materials of the two layers that the via connects, the hole through which connection is made, and metal that joint the two together.

From the layout mask point of view, we need a mask for the contact hole and the metal 1 layer covering the hole and other wires.

Mask set vs Fabricated circuit



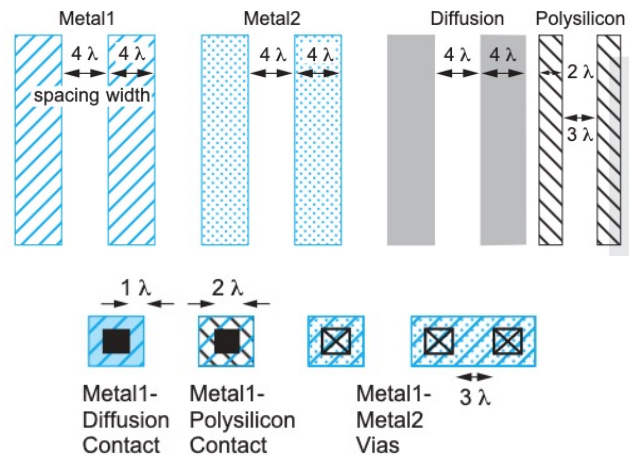
W&H p155

The upper diagram shows a typical mask set that is used to fabricate the inverter.

The lower diagram shows the actual polygons draw by the designer. While the two are similar, they are not the same. The tapeout process involve deriving the large set of masks required by the process to make the inverter from the layout design.

Scalable Layout Design Rules (1)

- ◆ Design Rules specify geometric constraints on the layout to guarantee functioning chips.
- ◆ Lambda λ is unit of measure often used to define the rules.
- ◆ Unusually half of the smallest feature size (e.g. for 180nm, $\lambda = 90\text{nm}$)



Design Rules specify geometric constraints on the layout to guarantee functioning chips.

There are two main types of rules:

1. Resolution constraints specify:

- Smallest width feature than can be used (e.g. channel length, wire width)
- Smallest spacing that will guarantee no shorts
- Governed by lithography and processing steps for technology used
- Resolution often depends on the smoothness of the surface/planarity

2. Alignment/overlap constraints specify:

- Alignment between layers
- Minimum overlaps or overhangs



The disadvantage of this is that the resulting rules tends to be conservative (having to round UP to the nearest multiple of λ). As a result, this it is not used for modern processes.

Advanced technologies have many additional rules including: poly/metal fill density rules, slotting rules, die corner rules, bond pad rules, antenna rules, ESD rules, and adaptive rules (e.g. the wider a metal trace – the wider the spacing).

Absolute Layout Design Rules

Layer / Feature	Rule	Typical Value (μm)	Notes
Poly (Gate)	Minimum width	0.07	Smallest gate length (L_{min})
Poly (Gate)	Minimum spacing	0.10	Between poly lines
Poly over Active	Extension	0.05	Poly overlap over diffusion edge
Active (Diffusion)	Minimum width	0.10	Smallest diffusion region
Active (Diffusion)	Minimum spacing	0.10	Between active regions
Contact (CA)	Contact size	0.08×0.08	Via/contact hole
Contact (CA)	Contact spacing	0.08	Edge-to-edge spacing
Metal 1	Minimum width	0.09	Local interconnect routing
Metal 1	Minimum spacing	0.09	Adjacent M1 spacing
Metal 2–6	Minimum width/spacing	0.10–0.14	Intermediate routing layers
Metal 7–9	Minimum width/spacing	0.28–0.50	Top-level and power routing
Via (V1–V8)	Size	0.08×0.08	Between metal layers
Via (V1–V8)	Spacing	0.08	Edge-to-edge spacing
N-Well / P-Well	Width / Spacing	0.40 / 0.40	For device isolation

This table shows a small collection of publicly available design rules for TSMC 65nm process.

The precise design rules for TSMC processes are proprietary and not publicly available. The numbers here are guidelines only. One has to consult the actual Design Rules Manual that is distributed alongside the PDK.

Those highlighted in GREEN are the rules relevant to Lab 2.

Some Layout Guidelines

MOSFET transistors

- Keep nMOS devices grouped together near VSS (can share P+ contacts)
- Keep pMOS devices grouped together near VDD (can share N+ contacts)
- Can place multiple PMOS devices within a single N-Well (if $V_B = V_S = V_{DD}$)

Interconnections (wires)

- Place pins (I/O connections) around perimeter of cell
- All long wires should use metal interconnections
- Route VDD and VSS in thick metal
- Interleave metal layers – i.e. M1/M3/M5 horizontal and M2/M4/M6 vertical
- Keep polysilicon/diffusion connections as short as possible (high resistivity)

General

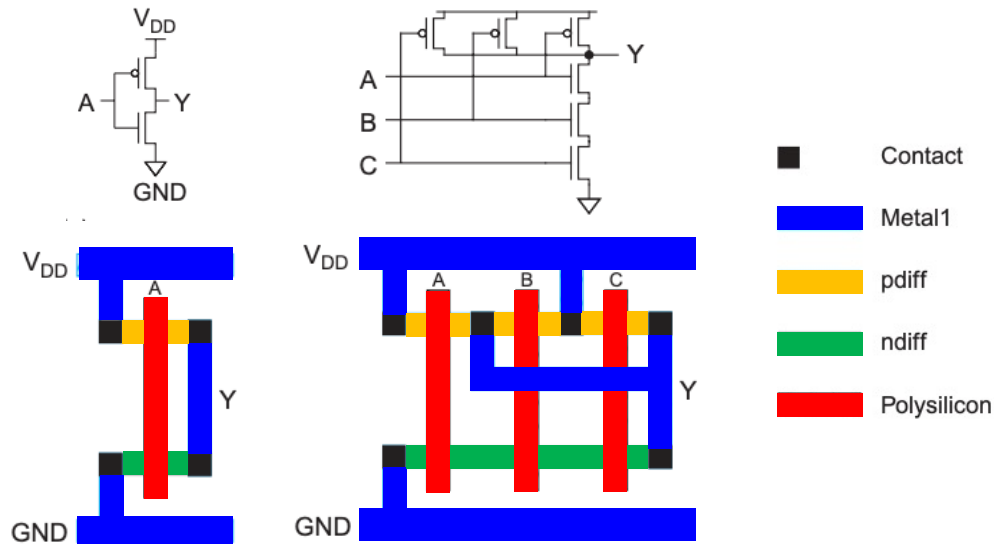
- Try to design/layout as regular as possible
- Plan your layout before hand

Source: Tim Constandinou

This slides are some simple guides that help you to create a good layout for your circuit. They are general rules that minimize potential errors and likely to result in a compact layout.

Planning your layout before hand is key. Without a clear plan, you may find that your layout is larger than it needs to be, and you may also introduce mistakes. You are almost certainly going to spend much longer to fix problems that you do not foresee.

Stick Diagram for Layout Design



Source: Tim Constandinou

One effective way to plan your layout is to use stick diagrams.

Stick diagrams are rough sketches of the layout using colour pen that ignores detail design rules. For example the layout of the inverter can be sketched quickly with the few lines showing the rough location of the pMOS and nMOS transistors. The contacts are just a black square or a cross. This can be transferred to the layout directly with ease.

The second circuit is more complex – a 3-input NAND gate. It shows how to three series nMOS transistors share their diffusion with neighbouring transistors. The topology of the pMOS transistors are also clearly show to minimize crossovers.

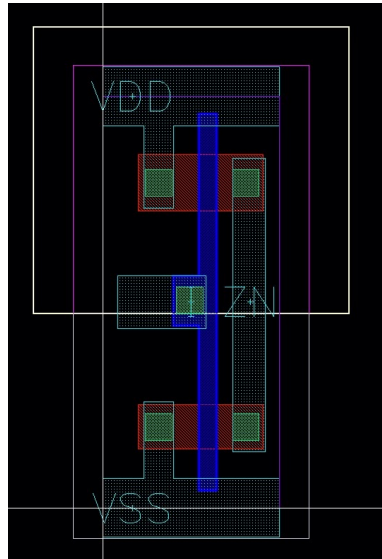
What to expect from Lab 2?

- ◆ Understand the different mask layers in a layout.
 - ◆ Read layout and “see” the circuit it implements.
 - ◆ Learn how to use the layout editor to wire things up.
 - ◆ Learn how to make sure that your design passes design rules (DRC) and the layout is the same as the schematic (LVS).
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- ◆ **Task 1:** Explore the layout of the inverter standard cell layer by layer.
 - ◆ **Task 2:** Extract transistor level circuit from XOR gate layout.
 - ◆ **Task 3:** Import circuit from Lab 1 and perform manual placement.
 - ◆ **Task 4:** Hand route the clock, control and data signals.
 - ◆ **Task 5:** Perform LVS verification.

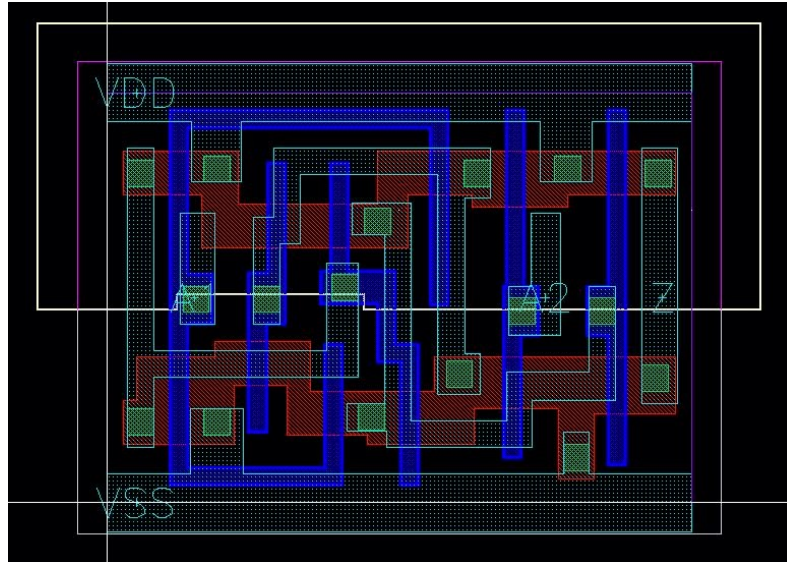
Here are four learning outcomes that Lab 2 is design to produce.

You will firstly explore the layout of the inverter standard cell, layer by layer. You will also learn how to “read” a layout and produce its equivalent transistor level schematic. You will then learn to hand place-and-route the circuit synthesized in Lab 1.

Task 1 – Understand layout of inverter



Task 2 – Extract circuit for XOR gate



Task 3 & 4 – Floorplan, Placement and Routing

